

Gender Differences in Epilepsy

*†Jakob Christensen, ‡§Marianne Juel Kjeldsen, †Henning Andersen, ‡Mogens Laue Friis,
and †Per Sidenius

**Department of Clinical Pharmacology, University of Aarhus, and †Department of Neurology, Aarhus University Hospital, Aarhus; and ‡Department of Neurology, Odense University Hospital, and §The Danish Twin Registry, Institute of Public Health, University of Southern Denmark, Odense University, Odense, Denmark*

Summary: *Purpose:* The aim of this study was to look at gender differences in unselected populations of patients with epilepsy classified according to the 1989 International League Against Epilepsy (ILAE) criteria.

Methods: Data were obtained from two sources: (a) the EpiBase database at the outpatient clinic at the Department of Neurology, Aarhus University Hospital, Denmark, confined to adults with epilepsy ($n = 2,170$), and (b) the Danish Twin Registry ($n = 318$).

Results: In localization-related epilepsy, no overall gender difference was found in either the EpiBase population ($n = 1,511$; $w = 750$ (50%), $m = 761$ (50%); $p = 0.80$) or in the twin population ($n = 172$; $w = 86$ (50%), $m = 86$ (50%); $p = 1.00$). However, in the EpiBase population, localization-related symptomatic epilepsies were more frequent in men ($n = 939$; $w = 426$ (45%), $m = 513$ (55%); $p = 0.005$); and cryptogenic localization-related epilepsies were more frequent in women (n

$= 572$; $w = 324$ (57%), $m = 248$ (43%); $p = 0.002$). In generalized epilepsy, more women than men were diagnosed in both populations [EpiBase: $n = 480$, $w = 280$ (58%), $m = 200$ (42%); $p < 0.001$; twin population: $n = 105$, $w = 63$ (60%), $m = 42$ (40%); $p = 0.05$]. The difference was confined to idiopathic generalized epilepsy [EpiBase: $n = 437$, $w = 259$ (59%), $m = 178$ (41%); $p < 0.001$; twin population: $n = 94$, $w = 60$ (64%), $m = 34$ (36%); $p = 0.01$].

Conclusions: More women than men were diagnosed with idiopathic generalized epilepsy in two epilepsy populations. Overall, no gender difference was found in localization-related epilepsy, but localization-related symptomatic epilepsies were more frequent in men, and cryptogenic localization-related epilepsies were more frequent in women. The results suggest a gender susceptibility to the development of specific epilepsy subtypes. **Key Words:** Gender difference—Generalized epilepsy—Classification—Epidemiology—Sex.

Epidemiologic studies of epilepsy indicate that the overall incidence of epilepsy is slightly higher in male than in female subjects (1,2). For the individual seizure types, various sex ratios have been reported (2–5). Hauser et al. reported population-based incidence cases in accordance with the 1981 International League Against Epilepsy (ILAE) criteria for seizure type (2,6). Age-adjusted incidence was higher in male than in female subjects for both generalized and partial seizures; however, the age-adjusted incidence of epilepsy characterized primarily by absence seizures at diagnosis was higher for female subjects (2). Several studies found a difference in sex ratio in selected groups of patients with primarily generalized epilepsies (3–5,7,8) and in the susceptibility of the brain to seizure-induced damage (9). However, we have been unable to locate any systematic study of epilepsy pop-

ulations addressing gender differences in epilepsy when classified according to 1989 ILAE criteria for epilepsies and epileptic syndromes (10). The aim of this study was to look at gender differences in unselected populations of patients with epilepsy.

METHODS

Data were obtained from two sources: (a) adult epilepsy patients from the outpatient clinic at the Department of Neurology, Aarhus University Hospital, Denmark; and (b) epilepsy patients from The Danish Twin Registry

At the outpatient clinic at Aarhus University Hospital, information on epilepsy-related issues for patients older than 15 years has been registered in a national epilepsy database since 1999 (EpiBase, Langved Data, Odense, Denmark). The database has information for each person on the specific epilepsy diagnosis according to 1989 ILAE criteria for epilepsies and epileptic syndromes (10). Furthermore, the database contains information on sex, age, use of antiepileptic drugs (AEDs), duration of disease,

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Address correspondence and reprint requests to Dr. J. Christensen at Department of Neurology, Aarhus University Hospital, Norrebrogade 44, DK-8000 Aarhus C, Denmark. E-mail: Jakob@farm.au.dk

presumed cause of epilepsy (if any), familial occurrence of seizures, and types and frequency of seizures. The data for this study were drawn from the database up to April 2003. Permission to use the database was granted by the Danish Data Protection Agency. On January 1, 2004, the number of persons older than 15 years living in Aarhus County, Denmark, was 527,773 (12% of the Danish population; www.statistikbanken.dk). Only one neurological department in the county provides care for adult patients (older than 15 years) free of charge. In Aarhus County, adult persons with epilepsy are referred for treatment mainly to the outpatient clinic at the Department of Neurology, Aarhus University Hospital. Private practicing physicians also provide care for persons with epilepsy (also free of charge), but this probably accounts for a minor proportion of patients. Gender distribution was assessed in a twin study based on data from the population-based Danish Twin Registry (11). This study evaluated self-reported epilepsy occurring among twins born between 1953 and 1982 (11). Seizures and epilepsies were classified according to ILAE criteria by extensive use of interview and medical records (10,11). The twin study was not confined to adults but evaluated all types of seizures and epilepsies in the twin population. For the purpose of this study, we excluded patients having only febrile seizures. Fewer than 2% of patients in EpiBase are expected to be included in The Danish Twin Registry.

TABLE 1. Gender differences according to 1989 ILAE-criteria. Results from EpiBase

Type of epilepsy	All epilepsies			Significance
	Women (%)	Men (%)	Total	
Localization-related epilepsy	750 (50)	761 (50)	1,511	p = 0.80
Generalized epilepsy	280 (58)	200 (42)	480	p < 0.001
Undetermined epilepsy	58 (41)	85 (59)	143	p = 0.03
Special syndromes	10 (28)	26 (72)	36	p = 0.01
All epilepsies (total)	1,098 (51)	1,072 (49)	2,170	p = 0.59

Patients with special syndromes included (male/female): stress convulsions (1/0); alcohol-induced seizures (20/2); other types of situation-related seizures (0/1); isolated, apparently unprovoked seizures (3/2); reflex epilepsy (2/3); and Rasmussen encephalitis (0/2).

The median age of the EpiBase population was 43.1 years. For the population aged 43 years in Aarhus County, the male/female ratio was 0.4985 (www.statistikbanken.dk). Use of a binomial test assuming a male/female ratio of 0.50 therefore assessed the gender difference. Data were analyzed by using SPSS (SPSS Inc, Chicago, IL, U.S.A.) version 11.0.

RESULTS

In April 2003, 2,170 persons were classified as having a diagnosis of epilepsy in the epilepsy database (EpiBase) at Aarhus University Hospital. This accounts for ~0.4% of

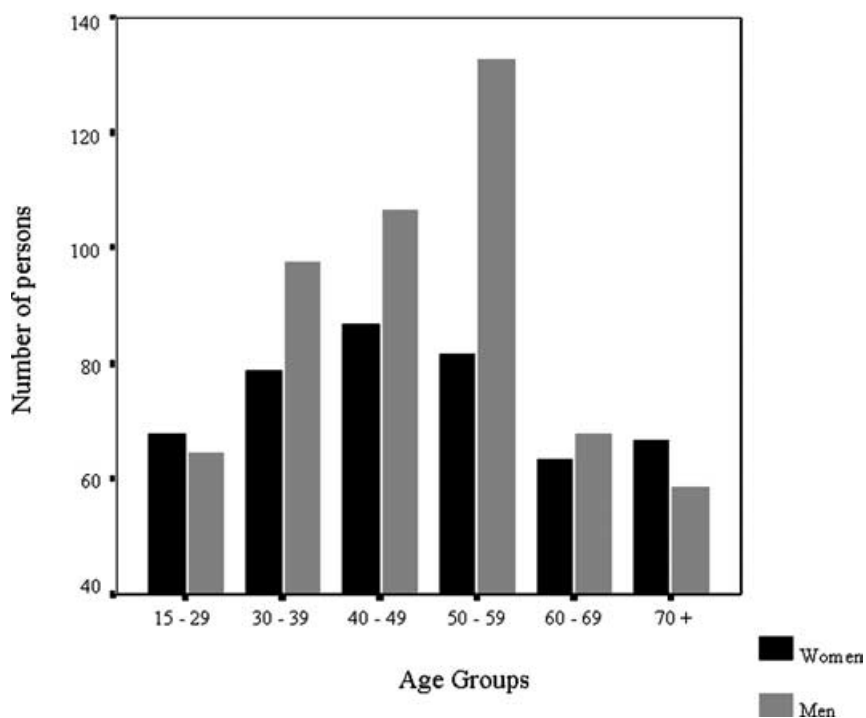
TABLE 2A. Gender differences in localization-related epilepsy from EpiBase, according to 1989 ILAE criteria

Type of epilepsy	Women (%)	Men (%)	Total	Significance
Symptomatic				
Frontal lobe epilepsy	70 (38)	116 (62)	186	p = 0.001
Temporal lobe epilepsies	156 (51)	152 (49)	308	p = 0.86
Parietal lobe epilepsies	34 (47)	38 (53)	72	p = 0.72
Occipital lobe epilepsies	13 (68)	6 (32)	19	p = 0.17
Other, multifocal or unknown localization	153 (43)	201 (57)	354	p = 0.01
All symptomatic	426 (45)	513 (55)	939	p = 0.005
Cryptogenic				
Frontal lobe epilepsy	26 (53)	23 (47)	49	p = 0.78
Temporal lobe epilepsies	180 (62)	109 (38)	289	p < 0.001
Parietal lobe epilepsies	11 (58)	8 (42)	19	p = 0.65
Occipital lobe epilepsies	7 (58)	5 (42)	12	p = 0.77
Other, multifocal or unknown localization	100 (49)	103 (51)	203	p = 0.88
All cryptogenic	324 (57)	248 (43)	572	p = 0.002
All localization-related epilepsies	750 (50)	761 (50)	1,511	p = 0.80

TABLE 2B. Gender differences in generalized epilepsy from EpiBase, according to 1989 ILAE criteria

Type of epilepsy	Women (%)	Men (%)	Total	Significance
Idiopathic				
Childhood absence epilepsy (pyknolepsy)	12 (63)	7 (37)	19	p = 0.36
Juvenile absence epilepsy	52 (76)	16 (24)	68	p < 0.001
Juvenile myoclonic epilepsy	105 (61)	67 (39)	172	p = 0.005
Epilepsies with grand mal on awakening	16 (52)	15 (48)	31	p = 1.00
Undetermined idiopathic generalized epilepsy	74 (50)	73 (50)	147	p = 1.00
Total idiopathic	259 (59)	178 (41)	437	p < 0.001
Symptomatic/Cryptogenic	21 (49)	22 (51)	43	p = 1.00
Total generalized	280 (58)	200 (42)	480	p < 0.001

(a) Symptomatic localization-related epilepsy



(b) Cryptogenic localization-related epilepsy

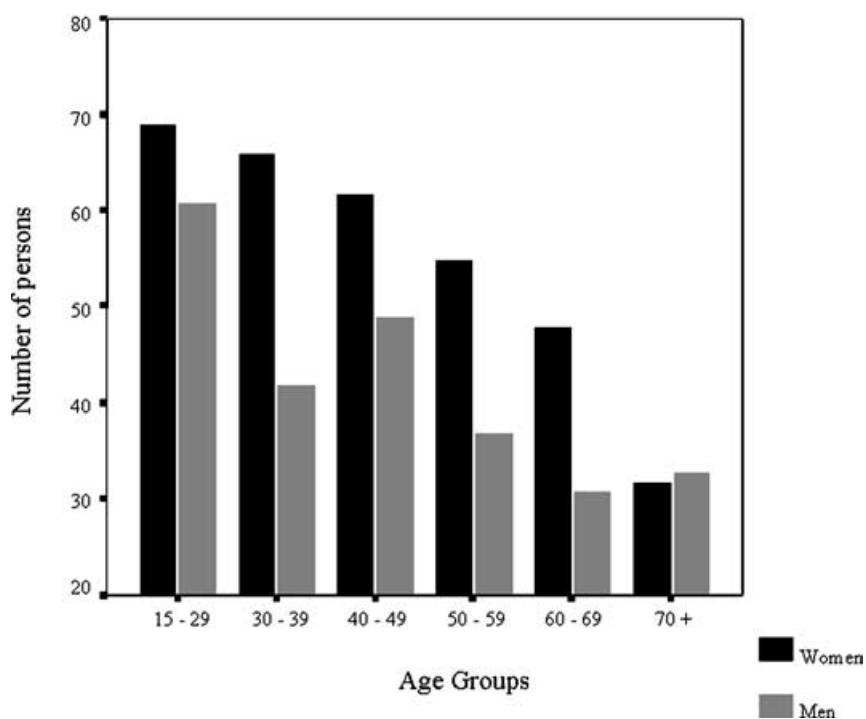


FIG. 1. Age distribution of patients for relevant epilepsy subtypes. Results from EpiBase. **A:** Localization-related symptomatic epilepsies ($n = 939$). **B:** Cryptogenic localization-related epilepsies ($n = 572$). **C:** Idiopathic generalized epilepsy ($n = 437$).

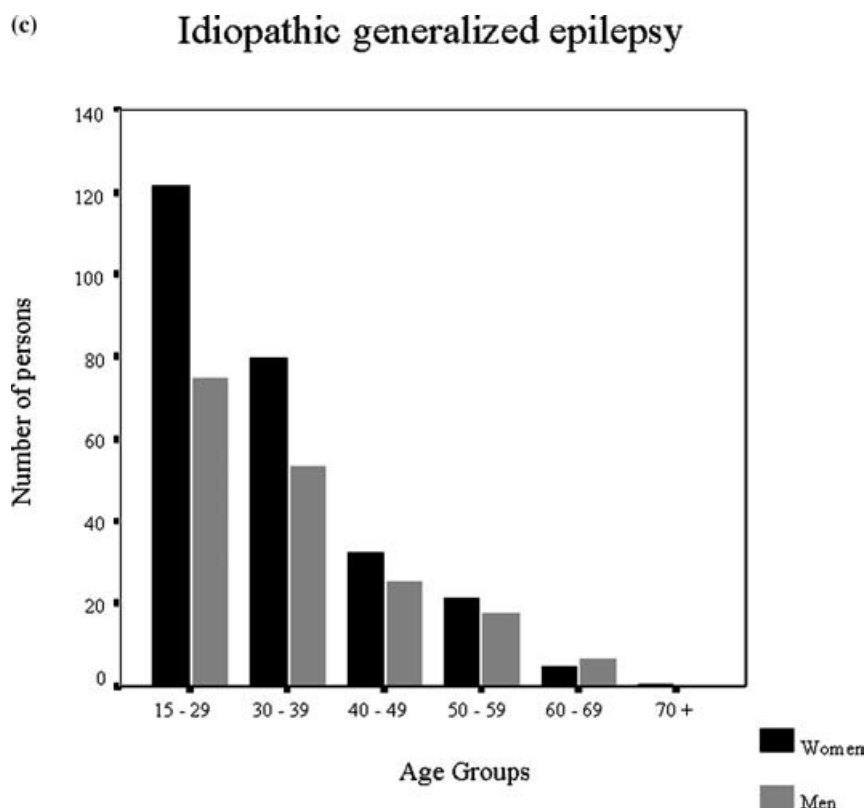


FIG. 1. Continued.

the population in Aarhus County. The gender distribution in relation to type of epilepsy is shown in Table 1. Overall, no gender difference was found, but generalized epilepsies were more frequent in women than in men. Further to explore the gender difference according to epilepsy type, we analyzed gender differences for localization-related and generalized epilepsies (Table 2A and B). Localization-related symptomatic epilepsies were more frequent in men than in women; however, cryptogenic localization-related epilepsies were more frequent in women (Table 2A). In patients with generalized epilepsies, the majority of patients were classified as having idiopathic generalized epilepsy. Idiopathic generalized epilepsy was more frequent in women than in men because of a gender difference in patients with juvenile absence epilepsy and juve-

nile myoclonic epilepsy (Table 2B). A gender difference also was observed for childhood absence epilepsy, but the numbers did not reach statistical significance.

We explored the gender differences for symptomatic and cryptogenic localization-related epilepsy and idiopathic generalized epilepsy according to age (Fig. 1). In symptomatic localization-related epilepsy, the gender difference was confined mainly to the age group 30–59 years; however, in cryptogenic localization-related epilepsy, the difference did not seem to be confined to any specific age group. In idiopathic generalized epilepsy, the gender difference declined with age.

In the population-based twin study of 318 validated cases, a gender difference was observed for idiopathic generalized epilepsy (Table 3). No statistically significant

TABLE 3. Gender differences from the Danish Twin Study, according to 1989 ILAE criteria

Type of epilepsy	Women (%)	Men (%)	Total	Significance
Localization-related epilepsy				
Idiopathic	7 (58)	5 (42)	12	$p = 0.77$
Symptomatic	29 (42)	40 (58)	69	$p = 0.23$
Cryptogenic	50 (55)	41 (45)	91	$p = 0.40$
Total localization-related epilepsy	86 (50)	86 (50)	172	$p = 1.00$
Generalized epilepsy				
Idiopathic	60 (64)	34 (36)	94	$p = 0.01$
Symptomatic/cryptogenic	3 (27)	8 (73)	11	$p = 0.23$
Total generalized epilepsy	63 (60)	42 (40)	105	$p = 0.05$
Undetermined epilepsy	3 (75)	1 (25)	4	$p = 0.25$
Special syndromes	18 (49)	19 (51)	37	$p = 0.26$
All epilepsies	170 (53)	148 (47)	318	$p = 0.24$

gender difference was noted in localization-related epilepsies, but a trend was observed for symptomatic cases to be more frequent in men and cryptogenic cases to be more frequent in women (Table 3). Because of the few cases in this population, it was not possible to examine further gender differences in detail.

DISCUSSION

In two population-based studies, we identified a gender difference in idiopathic generalized epilepsy. In the outpatient study (EpiBase), this difference was due to a gender difference in juvenile absence epilepsy and juvenile myoclonic epilepsy. A gender trend also was observed in childhood absence epilepsy, but only a few patients were assessed because only adults were included in this population. Other studies reporting gender differences in generalized epilepsies have found similar results (2–5,7,8), and in general, it seems that women more frequently than men have idiopathic generalized epilepsy. The reason behind this difference is not established, but it is likely that sex hormones may play a role in the development of idiopathic generalized epilepsy. If this assumption is true, the gender difference would be more pronounced before menopause, and indeed, the female preponderance in idiopathic generalized epilepsy was highest for the age group 15–50 years and declined with age.

Differences in age distribution between men and women could bias the results, but the age distributions of men and women were similar and thus probably they do not account for the differences observed.

Symptomatic localization-related epilepsy was more frequent among men than among women, which has been reported previously, and this may reflect differences in risk of structural damage to the brain and subsequent seizures (2,12). Accordingly, the gender difference was greatest in the age group 30–59 years (i.e., the age group with a high risk of traumatic brain injury). However, men also may be more vulnerable to seizure-associated brain damage (9), extent of brain involvement associated with seizures (13), and development of generalized tonic-clonic seizures (14). Among those with cryptogenic localization-related epilepsy, women were more frequent. In general it therefore seems that nonsymptomatic epilepsy (both idiopathic generalized and cryptogenic localization-related epilepsy) more often occurs in women in contrast to symptomatic localization-related epilepsy, which has a preponderance in men.

We were unable to exclude patients included in both cohorts, but we do not believe that this is a source of bias in the present study, because <2% of patients in EpiBase are expected also to be included in The Twin Registry.

The gender differences observed may be due to differences in ascertainment of patients. For example, women may be referred more often than men to treatment in the outpatient clinic, resulting in a “false” gender difference because of differences in referral patterns. The observed prevalence of epilepsy in our population was ~0.4%, which is probably lower than the “true” prevalence of epilepsy in the County of Aarhus (15) and indicates that not all patients have been referred for treatment to the Department of Neurology. We are therefore unable to exclude that a bias in referral pattern is the cause of the differences observed; however, it is striking that we found the same gender difference in two population-based groups of patients with epilepsy.

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